

Speculatively Ordered Reads for Byzantine Fault-Tolerant State-Machine Replication

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Abstract: State-machine replication is an established mechanism to build Byzantine fault-tolerant services, whereby a set of replicas redundantly process requests for the same application. In general, a consensus protocol is used to enforce that all replicas process client requests in the same order, which involves multiple communication steps. There is an established read-only optimization, which significantly reduces the latency for non-modifying requests by allowing them to bypass the consensus protocol. When enabled, this optimization requires to increase the response quorum sizes for both read and write requests to preserve linearizability. Further, the read-only optimization can result in a non-negligible read-write conflict rate even in the absence of Byzantine replicas. We propose the Speculatively-Ordered Read (SOR) optimization, which introduces one additional communication step where a chosen replica proposes execution orderings for read requests. We demonstrate that this approach drastically reduces the probability for read-write conflicts and thereby mitigates expensive request re-submissions.

Keywords: State-Machine Replication, Speculative Ordering, Read-Only Optimization

BFT consensus protocols [CL02, SB12, Yi19, Go19, ED21] involve multiple communication steps and enable replicas to agree on a common ordering of client requests. The required communication introduces significant overhead to the processing latency of client requests compared to non-replicated systems. Castro et al. [CL02] already proposed a read-only optimization as part of the PBFT protocol, in which replicas directly process read requests received from clients, since a possible out-of-order execution does not impact the consistency guarantees of the system. This optimization is able to significantly improve the end-to-end latency of read requests in wide-area deployments [SB15], especially for applications that primarily involve read requests. However, the read-only optimization requires clients to collect $2f + 1$ matching replies for both read and update requests to preserve linearizability [CL99]. Further, there can be situations in which clients are unable to retrieve the required number of matching responses. Such a situation arises if an interfering update request is concurrently ordered and delivered for execution to a subset of replicas before they process the read-only request and only afterwards to the other replicas.

In this talk, we will present the Speculatively-Ordered Read (SOR) optimization [HH26], which is able to drastically reduce the probability of read-write conflicts and thereby prevents unnecessary request re-submissions. The SOR optimization is based on an additional speculative ordering step for read-only requests, in which an execution position for a given request is proposed by a dedicated replica selected by the client. The approach

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is based on a simple heuristic which derives the earliest possible execution position that is consistently fulfillable by a quorum of replicas, relative to the current progress of the consensus protocol. We developed two variants of this heuristic, which provide a different trade-off between additionally introduced latency and compatibility with the established PBFT read-only optimization. We will present experimental results captured in different emulated wide-area deployments to demonstrate the feasibility of the approach.

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